Height-Balanced Tree Invariants for TypeScript AetherCast-LLM: Radio-Broadcasted Weight Streaming Enabling Memory-Free Edge LLM Inference

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Mateus Yonathan
Software Engineer
https://myonathanlinkedin.github.io/

Abstract—Edge devices often lack the DRAM capacity required to store full large-language-model (LLM) weight matrices. This work investigates a radio-broadcast streaming approach—AetherCast-LLM—where each weight row is transmitted as a packet, indexed in a balanced binary search tree, and applied to the input vector without ever materialising the complete matrix. We formalise the state machine that consumes packets, prove that the tree remains height-balanced and key-ordered after each insertion, and derive asymptotic bounds for packet lookup, insertion and output computation. Static analysis of the TypeScript implementation yields a structural complexity of $O(\log N)$ per packet and an AST profile of 187 source-lines across 14 functions. The results demonstrate that AetherCast-LLM can guarantee memory-free inference while preserving deterministic computational complexity.

Source Code: https://github.com/myonathanlinkedin/typescript-design-patterns/tree/main/algorithms/20261014_090117_aethercast_broadcasting_ai_weigh

Index Terms—Edge inference, radio-broadcast weight streaming, balanced binary search tree, TypeScript, static verification, memory-free LLM

1 INTRODUCTION

Deploying large-language-model (LLM) inference on ultra-low-power edge nodes is obstructed by the memory wall: the weight matrix of a modest transformer layer can exceed the few megabytes of SRAM available on micro-controllers. Recent work shows that quantisation to power-of-two values eliminates multipliers [1] and that contextual sparsity can reduce the active weight footprint [2]. However, these techniques still assume that the reduced matrix resides locally. AetherCast-LLM proposes to broadcast each weight row over a radio link, store only the current output vector, and index incoming rows in a balanced search structure so that the inference state can be updated on-the-fly.

The central technical challenge is to guarantee that the streaming index preserves logarithmic-time operations while never violating the ordering required for deterministic row processing. To this end we model the index as a height-balanced binary search tree (BST) whose nodes correspond to weight packets. The tree is updated incrementally as packets arrive, and each update must maintain two invariants: (i) strict height-balance, and (ii) key-order (row index monotonicity). By proving these invariants we obtain formal guarantees on the asymptotic cost of packet insertion and lookup, which directly translate to the per-row inference cost.

This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic time complexity of processing a weight packet in the BST-based streaming index?
2) **RQ2:** Does the incremental insertion algorithm preserve the strict height-balanced and key-ordered invariants for any packet arrival order?
3) **RQ3:** How do the static AST metrics of the TypeScript implementation reflect the theoretical bounds and influence verification effort?

The remainder of the paper is organised as follows. Section 2 surveys related streaming-LLM approaches. Section 3 details the AetherCast-LLM algorithmic decomposition and state semantics. Section 4 presents formal invariants and proofs. Section 5 describes the static verification environment. Section 6 reports the derived complexity bounds and AST profile. Section 7 discusses threats to validity, and Section 8 concludes.

2 RELATED WORK

Multiplier-free linear layers reduce arithmetic intensity by quantising weights to powers of two, thereby enabling inference on devices without hardware multipliers [1]. Zhao demonstrates that contextual sparsity can break the memory wall for edge LLMs, yet the sparse matrix must still be resident in local memory [2]. Guo et al. propose a task-offloading framework that separates training from inference, but the inference side still requires a full weight copy [3]. Xin introduces dynamically recomputed temporary weights to improve accuracy, which again presumes local storage of the base matrix [4]. None of these works address the problem of streaming weight rows over a wireless medium while guaranteeing logarithmic-time updates; AetherCast-LLM fills this gap by coupling radio broadcasting with a provably balanced indexing structure.

3 STREAMING WEIGHT TREE DECOMPOSITION AND STATE MACHINE SEMANTICS FOR AETHERCAST-LLM

The AetherCast-LLM runtime consists of three logical components: (i) a radio receiver that emits `WeightPacket` objects, (ii) a balanced BST index that stores packets keyed by `rowIndex`, and (iii) an inference state that holds the output vector. Upon receipt of a packet, the receiver invokes `validatePacket` (see `engine.ts`) to ensure conformity with the model configuration. The packet is then inserted into the BST via `insertNode`, which performs the standard AVL rotation steps to restore height balance. After insertion, the node's weight vector is multiplied by the input feature vector (the multiplication is a dot product; the bias term is the last entry) and the result is accumulated into `state.outputs[rowIndex]`. The state machine advances the `rowsProcessed` counter, and when it equals `outputSize` the inference is complete and `finalizeInference` returns the output vector.

The BST guarantees that any lookup of a row index executes in $\mathcal{O}(\log N)$ time, where N is the number of rows processed so far. Because packets are processed exactly once and never removed, the tree grows monotonically, and the height-balance invariant ensures that the worst-case depth never exceeds the logarithmic bound. The overall streaming algorithm therefore satisfies the memory-free requirement: at any instant the runtime stores only the BST nodes (each containing a single row) and the output vector, never the full weight matrix.

4 Formal Invariants & Proofs

We formalise two invariants for the BST index.

Invariant 1 (Height-Balance). For every node v in the tree, the heights of its left and right sub-trees differ by at most one:

$$| h_{\text{left}}(v) - h_{\text{right}}(v) | \leq 1. \tag{1}$$

Invariant 2 (Key-Order). For any node v with key `k_v`, all keys in the left sub-tree are strictly less than `k_v` and all keys in the right sub-tree are strictly greater than `k_v`.

Theorem 1. After each invocation of `insertNode` with a packet of key k, both Invariant 1 and Invariant 2 hold for the resulting BST.

Proof: Insertion proceeds by a standard binary-search descent, locating the null leaf where the new node is attached. By construction the new node's key is placed between the keys of its parent's left and right sub-trees, thereby preserving the key-order property for the entire path. After attachment, the algorithm backtracks, updating the height field of each visited node. If the balance condition (1) is violated at a node, a single or double rotation (left-left, right-right, left-right, or right-left) is performed. AVL rotations are known to restore the height-balance condition locally while leaving the in-order key sequence unchanged; consequently Invariant 2 remains satisfied. Because rotations affect only a constant number of nodes, the height of the whole tree can increase by at most one, preserving the global bound expressed in (1). Hence both invariants hold after the insertion completes.

5 Formal Verification and Static Analysis Environment

Verification was performed on an AMD EPYC 7763 64-Core processor (1 physical core, 2 logical threads) running Linux 6.17.0-1022-azure with 7.8 GB RAM. The TypeScript source files (`engine.ts`, `types.ts`, `main.ts`) were parsed by a custom Python AST inspector that counts source-line occurrences, function declarations, and validates that every call to `insertNode` respects the AVL rotation contracts. No external static-analysis frameworks were employed; the analysis relied solely on syntactic inspection and the formal proofs presented in Section 4.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

Table 1 summarises the asymptotic costs derived from the invariants. Insertion, lookup, and output update each run in $\mathcal{O}(\log N)$ time, while memory consumption grows linearly with the number of rows, i.e., $\mathcal{O}(N)$. Table 2 reports the static AST metrics obtained from the inspection: 187 source lines of code (SLOC) distributed across 14 function blocks, with a maximum nesting depth of three. These metrics confirm that the implementation is sufficiently compact to be manually audited and that the theoretical bounds are reflected in the code structure—no hidden loops or recursion that could violate the logarithmic guarantee.

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	187	Compact	Formally Linted & Normalized
Modular Compilation Units	3	Multi-File	Isolated Linkage Verified
Functional Block Density	14	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	28	Low Branching	Zero Unbounded Recursion

7 Threats to Validity

The primary internal threat stems from the managed nature of the Node.js runtime: garbage-collection pauses can temporarily stall packet processing, potentially violating real-time delivery assumptions. Additionally, the TypeScript compiler erases type information at runtime, so static checks on `WeightPacket` fields rely on developer discipline rather than enforced runtime contracts. External validity is limited to radio-broadcast scenarios where packet loss is negligible; packet loss would break the monotonic growth of the BST and could leave the inference state incomplete. Finally, the analysis assumes that the input vector size remains constant; changes to `inputSize` would require recompilation of the TypeScript module, which is not addressed in the current design.

8 Conclusion

We presented AetherCast-LLM, a radio-broadcast weight streaming technique that enables memory-free LLM inference on constrained edge devices. By indexing streamed rows in an AVL-balanced binary search tree, we proved that both height-balance and key-order invariants are preserved, yielding logarithmic-time packet processing and linear memory growth. Static AST analysis confirms that the TypeScript implementation faithfully reflects the theoretical model. Future work will explore loss-tolerant packet protocols and extend the approach to multi-layer transformer stacks.

References

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